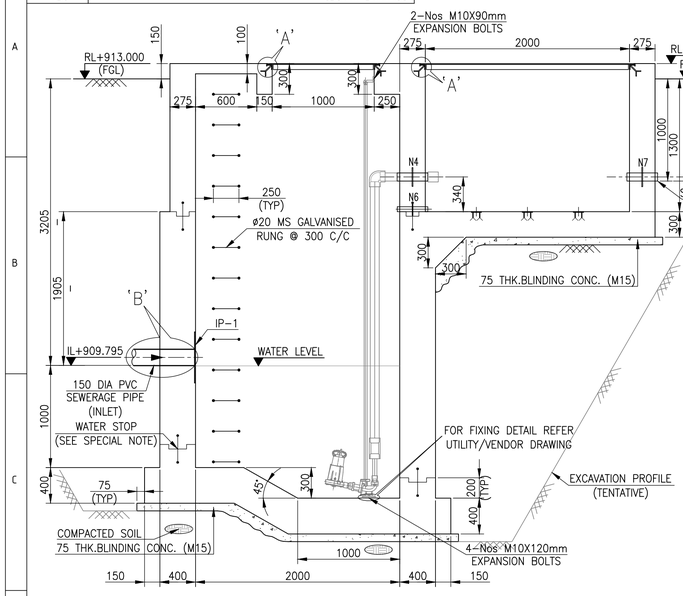
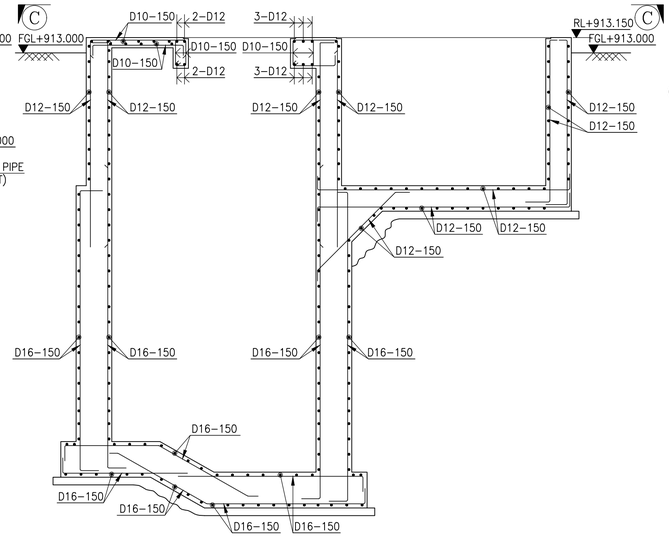
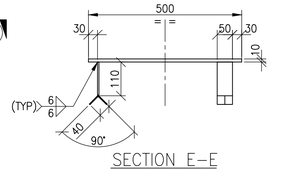
DRAWING NUMBER: MD-216-49EE-EG-CV-GAD-1001 | REV: D01



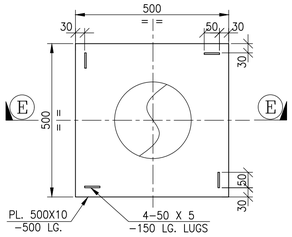
SECTION A–A
SCALE= 1:30



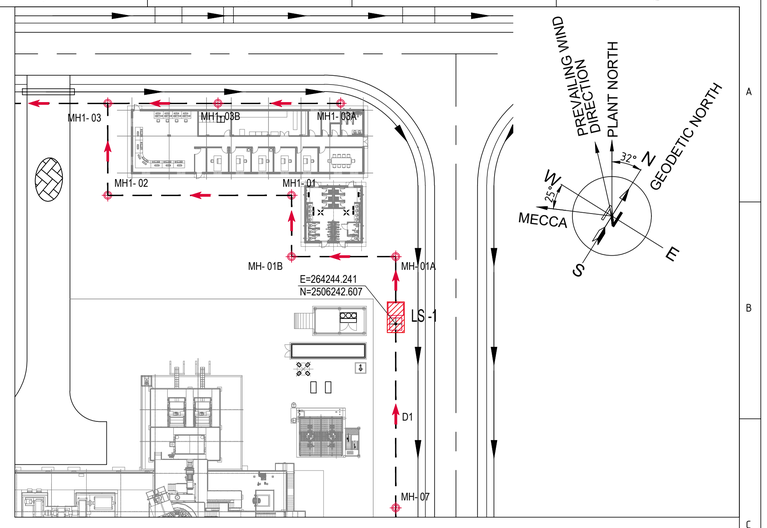
SECTION B–B
SCALE= 1:30



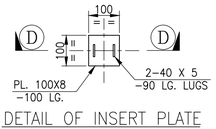
SECTION E–E
SCALE= 1:10



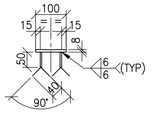
DETAIL OF INSERT PLATE MKD. IP-1
SCALE= 1:10



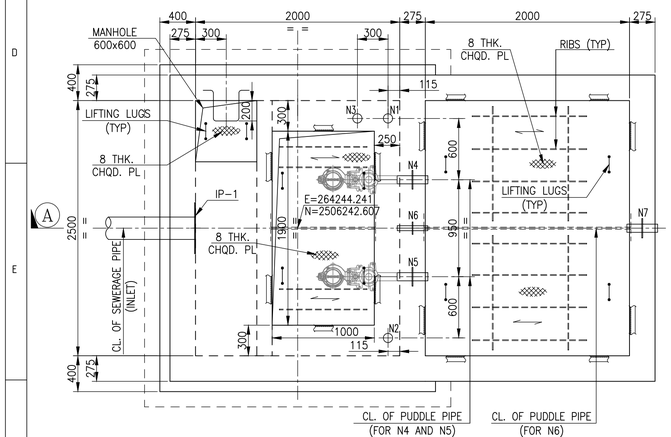
KEY PLAN
SCALE – 1:500



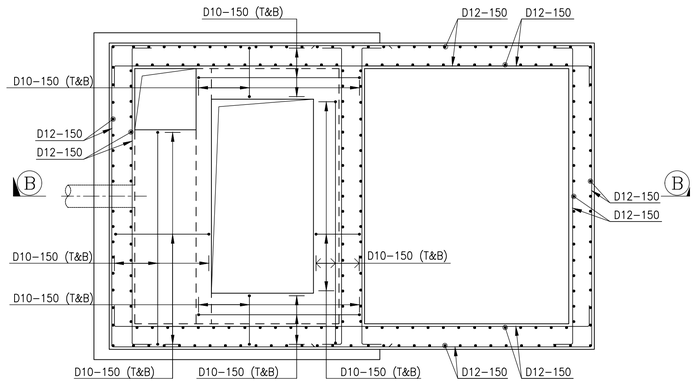
DETAIL OF INSERT PLATE MKD. IP-2
SCALE= 1:10



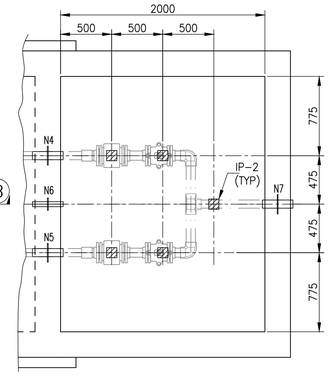
SECTION D–D
SCALE= 1:10



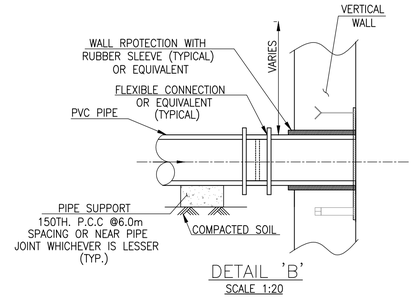
TOP PLAN & DETAILS OF LIFTING STATION–1
SCALE= 1:30



PLAN C–C
R.C DETAIL OF LS–1
SCALE= 1:30



PLAN AT EL+912.000
SCALE= 1:30



DETAIL 'B'
SCALE 1:20

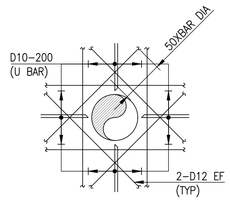
LEGENDS:

TYP – TYPICAL
RL – REDUCED LEVEL
IL – INVERT LEVEL
FGL – FINISHED GROUND LEVEL
R.C – REINFORCED CONCRETE
DIA – DIAMETER
THK – THICK
C/L – CENTRE LINE
P.C.C – PLAIN CEMENT CONCRETE
R.C.C – REINFORCED CEMENT CONCRETE
MS – MILD STEEL
PVC – POLYVINYL CHLORIDE
PL – PLATE
LG. – LONG
MKD. – MARKED
CONC. – CONCRETE
T&B – TOP AND BOTTOM
CHQD. – CHEQUERED
LS – LIFTING STATION
RENIF. – REINFORCEMENT

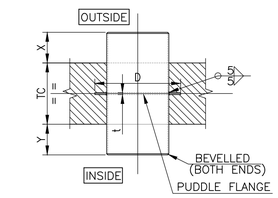
PIPE SLEEVE SCHEDULE

MARKING OF PIPE	DESCRIPTION	NO.	'X' (MM)	'TC' (MM)	'Y' (MM)	'D' (MM)	't' (MM)
N1	PUMP POWER CABLE	1	–	300	100	DN80	12
N2	PUMP CONTROL CABLE	1	–	300	100	DN80	12
N3	FLOAT SWITCH	1	–	100	100	DN80	12
N4	PUMP OUT	1	100	275	100	DN80	12
N5	PUMP OUT	1	100	275	100	DN80	12
N6	CHAMBER DRAIN	1	–	275	–	DN40	12
N7	SEWAGE OUTLET	1	100	275	100	DN80	12

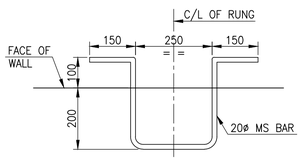
SPECIAL NOTE :
AT CONSTRUCTION JOINT WATER TIGHTNESS SHALL BE ENSURED BY PROVISION OF WATER STOP AS SUITABLE.



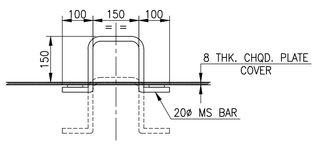
TYP. RENIF. DETAIL OF PIPE OPENING
SCALE= 1:20



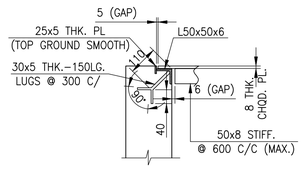
TYP. DETAIL OF PIPE SLEEVE
(FOR X, TC, Y, D & t REFER TABLE)
SCALE= 1:20



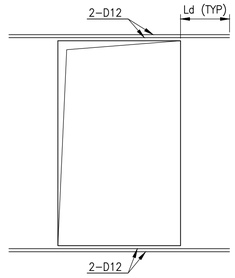
DETAIL OF RUNG
SCALE 1:10



DETAIL OF LIFTING LUG
SCALE 1:10



DETAIL 'A'
SCALE= 1:10



TYP. REINFORCEMENT DETAIL OF MANHOLE OPENING

NOTES:

1. ALL DIMENSIONS ARE IN mm AND LEVELS ARE IN METRE.
2. ALL CO–ORDINATES ARE IN METRES.
3. CO–ORDINATES ARE ON WORLD GRID.
4. FINISHED GROUND LEVEL (F.G.L.) IS BASED ON MEAN SEA LEVEL.
5. GRADE OF REINFORCED CEMENT CONCRETE (R.C.C) SHALL BE M35.
6. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
7. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
   a) WALL = 50 mm.
   b) SLAB = 25 mm.
8. BACKFILLING SHALL BE DONE SIMULTANEOUSLY AT ALL SIDES OF PIT.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	16/02/2023	AS-BULT	PML	BJB	APEX	SJM	
		C02	04/05/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 4A	PML	BJB	APEX	SJM	
MD-216-49EE-EG-CV-DED-1004	SCHEDULE OF SEWERAGE NETWORK MANHOLE	C01	02/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 4A	APEX	BJB	APEX	SJM	
MD-216-49EE-EG-CV-DED-1001	LAYOUT & DETAILS OF SEWERAGE NETWORK	A01	03/11/2020	ISSUED FOR REVIEW	APEX	BJB	APEX	SJM	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_910_C_V_24022_001 | REVISION: 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG90__MTL01__50002 | DOCUMENT ID: OU100673207 | REVISION: 3

LARSEN & TOUBRO
This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWN	APEX	30/10/2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	BJB	02/11/2020	GENERAL ARRANGEMENT
ENGINEER	APEX	28/10/2020	PROCESS PLANT SEWERAGE NETWORK
APPROVED	SJM	03/11/2020	PLAN, SECTION & DETAILS FOR LIFTING STATION–1
EMPLOYER			
SCALE	1:10, 20, 30, 500	A1	DRAWING NUMBER: MD-216-49EE-EG-CV-GAD-1001 | REV: D01